

Prime Factorization

Flagship

Lesson 1-1-flagship

Name:

Date:

Class:

ORBITAL LOGISTICS MISSION

Cargo Codebreak

You are the logistics officer aboard Station Helios. A shipment of 60 supply crates just docked, and the sorting robots can only distribute cargo once it is broken into its prime building blocks. Master prime factorization and the station eats this week.



TODAY'S OBJECTIVES

Content Objective: I can write a number as a product of its prime factors using a factor tree.

Language Objective: I can explain how I broke a number down using the words prime number, composite number, factor, and exponent.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Prime number

Composite number

Prime factorization

Factor tree

Exponent

Tap any word to see what it means and a picture.

1 A whole number greater than 1 with exactly two factors, 1 and itself, is a number.

2 A whole number greater than 1 that has more than two factors is a number.

3 — Writing a number as prime numbers multiplied together.

4 I can break a number into its prime factors step by step using a .

5 In 2^3 , the small number 3 is the and it shows 2 is multiplied 3 times.

Reflect — Exit Ticket

What is the prime factorization of 40?

A. $2 \times 2 \times 2 \times 5$

B. 4×10

C. 5×8

D. 2×20

Your answer:
